

Lecture 6: Joint Distributions

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Outline

- 1 Discrete Multivariate R.V.s
- 2 Continuous Multivariate R.V.s
- 3 Covariance and Correlation
- 4 Multinomial Distribution
- 5 Multivariate Normal

Multivariate Distribution

- Joint distribution provides complete information about how multiple r.v.s interact in high-dimensional space.
- Marginal distribution is the individual distribution of each r.v.
- Conditional distribution is the updated distribution for some r.v.s after observing other r.v.s.

Outline

- 1 Discrete Multivariate R.V.s
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Joint CDF

Definition

The *joint CDF* of r.v.s X and Y is the function $F_{X,Y}$ given by

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

and

The joint CDF of n r.v.s is defined analogously.

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

Joint PMF

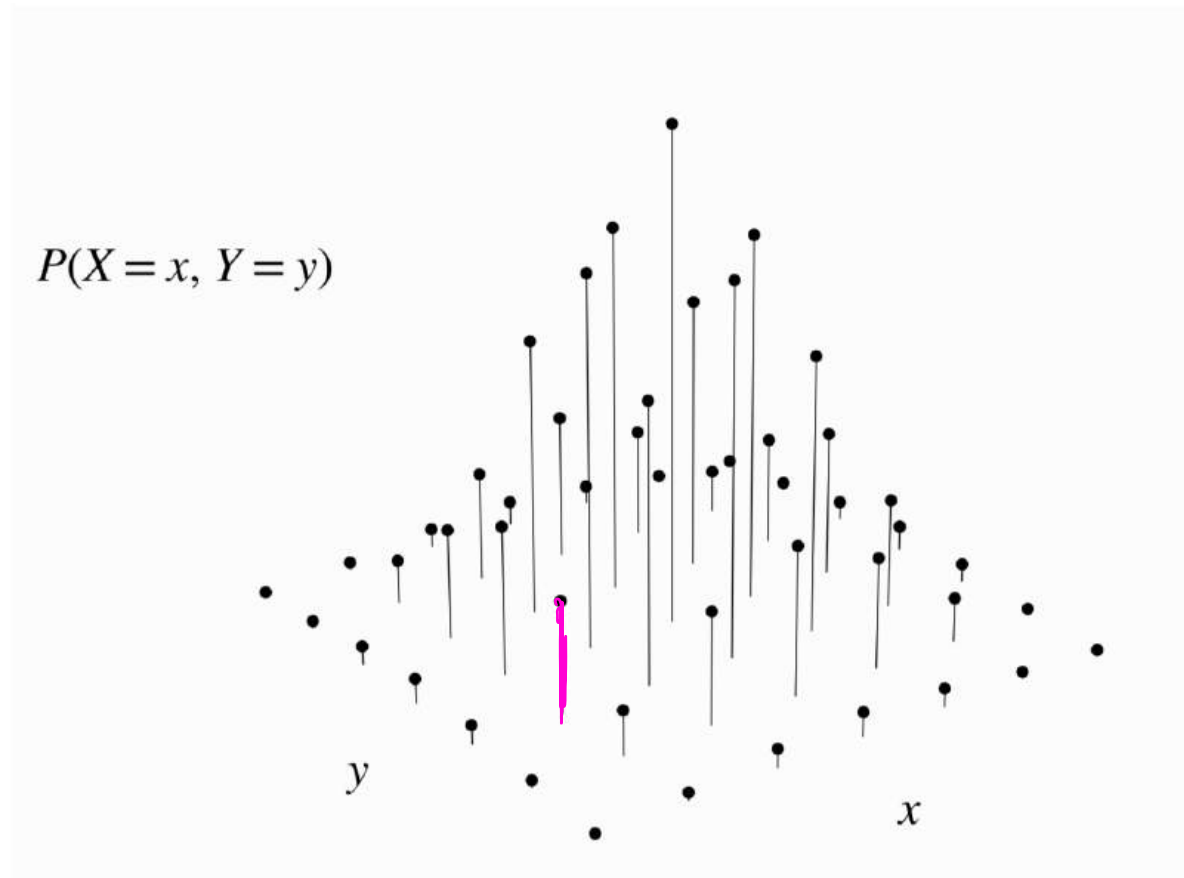
Definition

The joint PMF of discrete r.v.s X and Y is the function $p_{X,Y}$ given by

$$\underline{p_{X,Y}(x,y) = P(X = x, Y = y).}$$

The joint PMF of n discrete r.v.s is defined analogously.

Joint PMF



Marginal PMF

$$P(X=x) = \sum_y P(X=x|Y=y) P(Y=y)$$

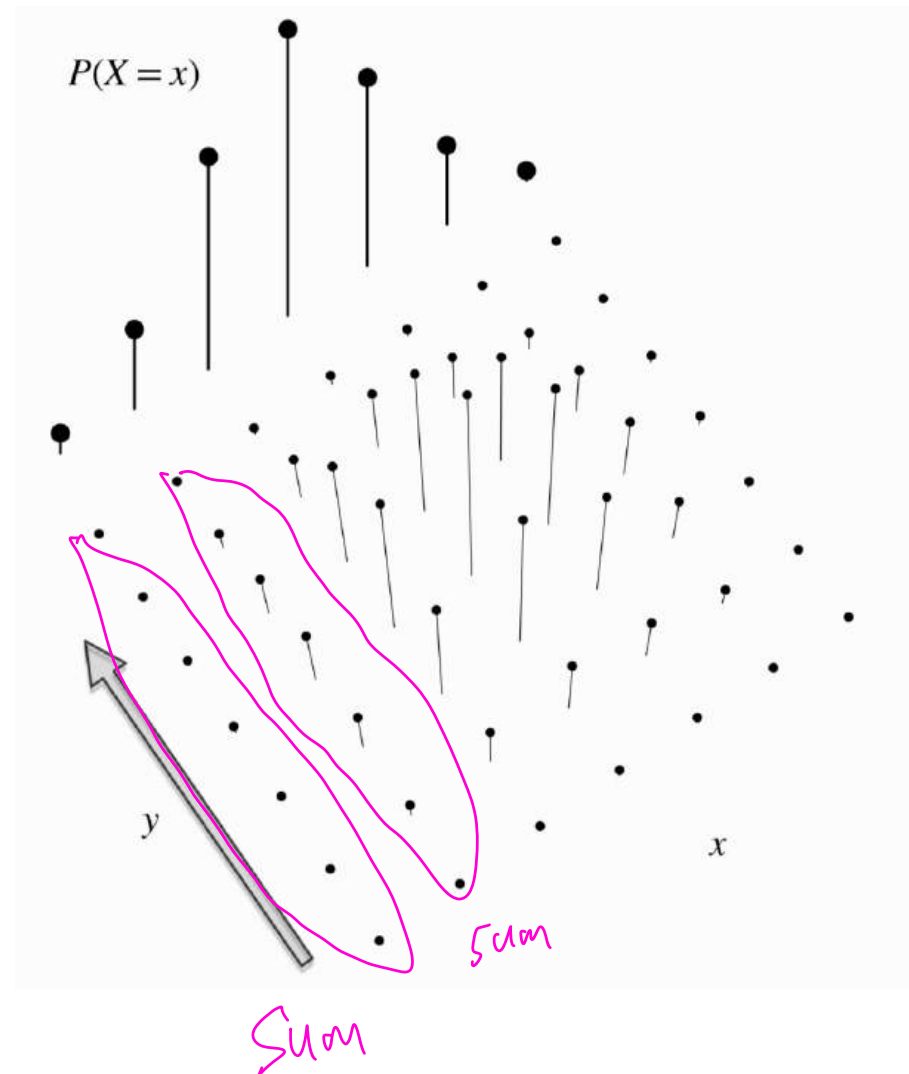
Definition

For discrete r.v.s X and Y , the *marginal PMF* of X is

$$P(X = x) = \sum_y P(X = x, Y = y).$$

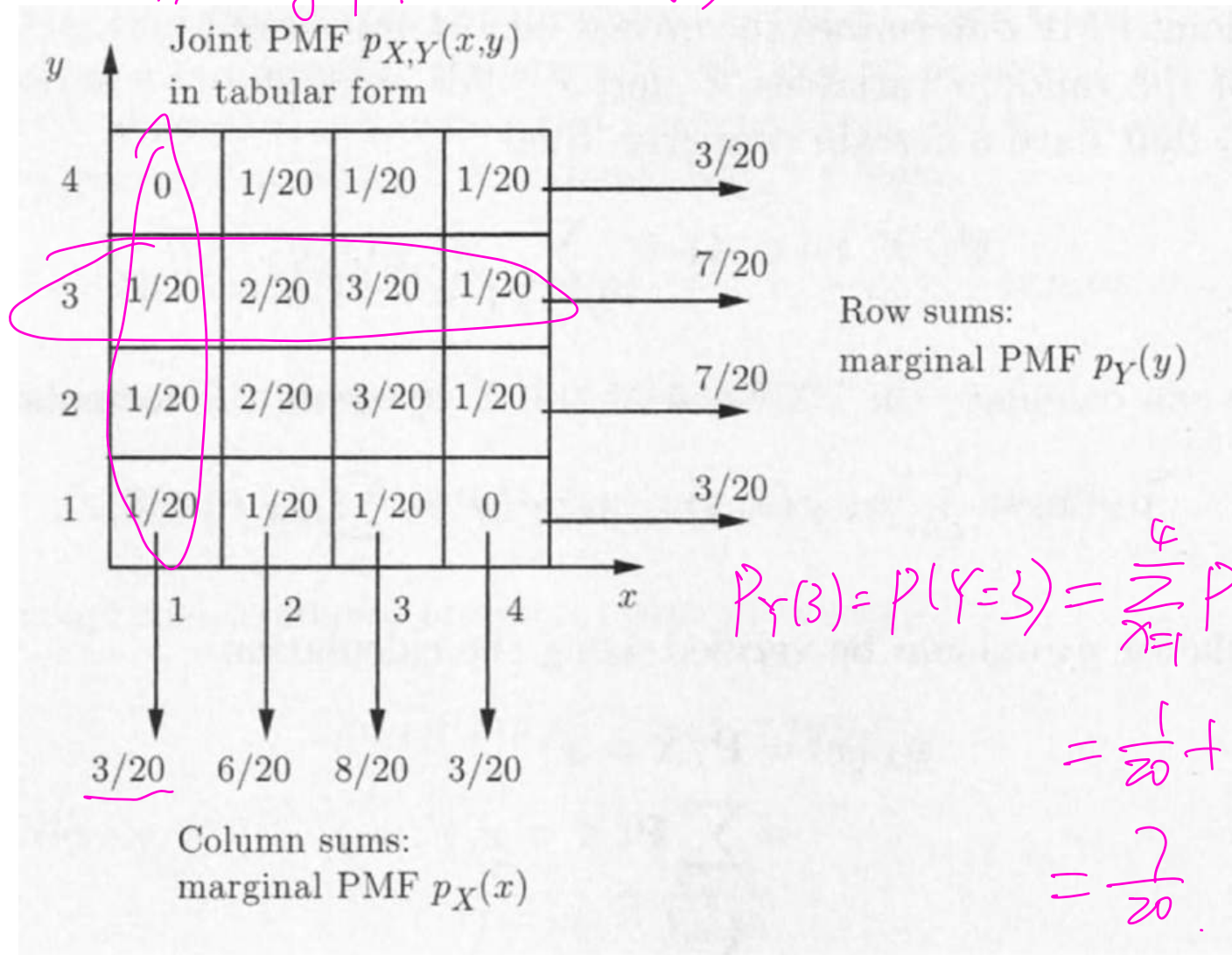
Marginal PMF

$P(X=x)$: total height of the bars
in the corresponding
column of the joint
PMF.



Example

$$P_X(1) = P(X=1) = \sum_{y=1}^4 P(X=1, Y=y) = \frac{1}{20} + \frac{1}{20} + \frac{1}{20} + 0 = \frac{3}{20}$$



$$P_Y(3) = P(Y=3) = \sum_{x=1}^4 P(X=x, Y=3)$$

$$= \frac{1}{20} + \frac{2}{20} + \frac{3}{20} + \frac{1}{20}$$

$$= \frac{7}{20}$$

Conditional PMF

∴ conditional PMF is also a valid PMF. $\begin{cases} \sum_x P_{X|Y}(x|y) = 1. \\ P_{X|Y}(x|y) \geq 0 \end{cases}$

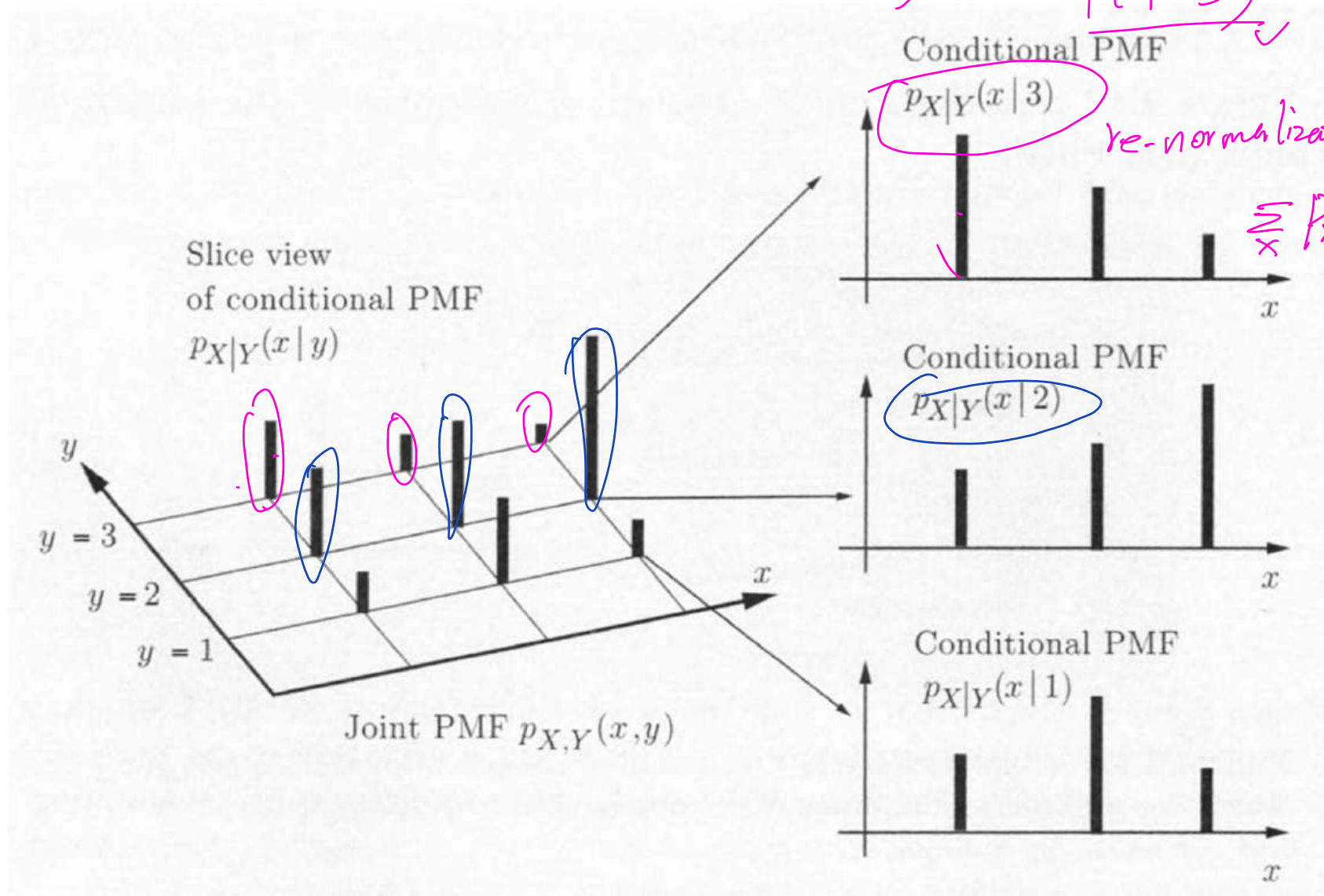
Definition

For discrete r.v.s X and Y , the *conditional PMF* of X given $Y = y$ is

$$P_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}.$$

Conditional PMF

$$p_{X|Y}(x|3) = \frac{p(X=x, Y=3)}{p(Y=3)}$$



Independence of Discrete R.V.s

Definition

Random variables X and Y are *independent* if for all x and y ,

$$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y), \forall x, y.$$

$$F_{X,Y}(x, y) = F_X(x) F_Y(y).$$

If X and Y are discrete, this is equivalent to the condition

$$P(X = x, Y = y) = P(X = x) P(Y = y)$$

for all x and y , and it is also equivalent to the condition

$$P(Y = y | X = x) = P(Y = y)$$

for all y and all x such that $P(X = x) > 0$.

Example: Chicken-egg

- ① Joint PMF $P(X=i, Y=j), i \in \mathbb{N}, j \in \mathbb{N}$.
- ② Conditioning on $N=n$. $X|N=n \sim \text{Bin}(n, p)$. $Y|N=n \sim \text{Bin}(n, q)$
 $X+Y|N=n = n$.

Suppose a chicken lays a random number of eggs, N , where $N \sim \text{Pois}(\lambda)$. Each egg independently hatches with probability p and fails to hatch with probability $q = 1 - p$. Let X be the number of eggs that hatch and Y the number that do not hatch, so $X + Y = N$. What is the joint PMF of X and Y ?

③ LOTP.
$$P(X=i, Y=j) = \sum_{n=0}^{\infty} P(X=i, Y=j, N=n) = P(X=i, Y=j, N=i+j)$$

$$P(A, B|C)$$

$$= P(A|C) P(B|A, C)$$

$$= P(X=i, Y=j | N=i+j) P(N=i+j)$$

$$= P(X=i | N=i+j) \underbrace{P(Y=j | N=i+j, X=i)}_1 P(N=i+j)$$

Solution

$$e^{-\lambda} = e^{-\lambda(p+q)} = e^{-\lambda p} \cdot e^{-\lambda q}$$

$$X|N=i+j \sim \text{Bin}(i+j, p)$$

$$N \sim \text{Pois}(\lambda)$$

$$P(X=i, Y=j) = P(X=i | N=i+j) P(N=i+j)$$

$$= \binom{i+j}{i} p^i q^j \frac{e^{-\lambda} \lambda^{i+j}}{(i+j)!} = \frac{\cancel{(i+j)!}}{i! j!} \cdot p^i q^j \cdot \frac{e^{-\lambda} \lambda^{i+j}}{\cancel{(i+j)!}}$$

$$= e^{-\lambda} \cdot \frac{(\lambda p)^i}{i!} \cdot \frac{(\lambda q)^j}{j!} = \frac{e^{-\lambda p} (\lambda p)^i}{i!} \cdot \frac{e^{-\lambda q} (\lambda q)^j}{j!}$$

$\sim \text{Pois}(\lambda p)$
 $\sim \text{Pois}(\lambda q)$

$$\textcircled{4} \quad P(X=i) = \sum_{j=0}^{\infty} P(X=i, Y=j) = \frac{e^{-\lambda p} (\lambda p)^i}{i!} \left(\sum_{j=0}^{\infty} \frac{e^{-\lambda q} (\lambda q)^j}{j!} \right)$$

$X \sim \text{Pois}(\lambda p)$
 $Y \sim \text{Pois}(\lambda q)$

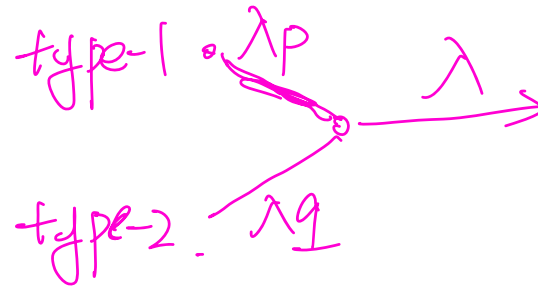
Solution

$$\textcircled{5} \quad P(X=i, Y=j) = P(X=i) P(Y=j), \quad \forall i, j \in \mathbb{N}^2$$

X, Y are independent. $(X+Y=N, N \sim \text{Pois}(\lambda))$

Conditioning on $N=n$, $X+Y=n$, X, Y not independent.

Related Theorem



Theorem

If $X \sim \text{Pois}(\lambda p)$, $Y \sim \text{Pois}(\lambda q)$, and X and Y are independent, then $N = X + Y \sim \text{Pois}(\lambda)$ and $X|N = n \sim \text{Bin}(n, p)$. p+q=1.

Related Theorem

Theorem

If $N \sim \text{Pois}(\lambda)$ and $X|N = n \sim \text{Bin}(n, p)$, then $X \sim \text{Pois}(\lambda p)$, $Y = N - X \sim \text{Pois}(\lambda q)$, and X and Y are independent.

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Conditional PDF Given an Event

Conditional PDF Given an Event

- The conditional PDF $f_{X|A}$ of a continuous random variable X , given an event A with $\mathbf{P}(A) > 0$, satisfies

$$\mathbf{P}(X \in B | A) = \int_B f_{X|A}(x) dx.$$

- If A is a subset of the real line with $\mathbf{P}(X \in A) > 0$, then

$$f_{X|\{X \in A\}}(x) = \begin{cases} \frac{f_X(x)}{\mathbf{P}(X \in A)}, & \text{if } x \in A, \\ 0, & \text{otherwise.} \end{cases}$$

- Let $A_1, A_2, \dots, A_n$ be disjoint events that form a partition of the sample space, and assume that $\mathbf{P}(A_i) > 0$ for all i . Then,

$$f_X(x) = \sum_{i=1}^n \mathbf{P}(A_i) f_{X|A_i}(x)$$

(a version of the total probability theorem).

Proof

$$\text{LOTP: } P(X \leq x) = \sum_{i=1}^n P(X \leq x | A_i) \cdot P(A_i)$$

$$P(X \leq x | A_i) = \int_{-\infty}^x f_{X|A_i}(t) dt.$$

$$F_X(x) = \underline{P(X \leq x)} = \sum_{i=1}^n P(A_i) \cdot \int_{-\infty}^x f_{X|A_i}(t) dt.$$

Taking derivatives of both sides

$$f_X(x) = \sum_{i=1}^n P(A_i) f_{X|A_i}(x).$$

Joint PDF

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

① Valid PDF $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx dy = 1. \quad f_{X,Y}(x,y) \geq 0.$

② Example: $P(X < 3, 1 \leq Y < 4) = \int_{-\infty}^3 \int_1^4 f_{X,Y}(x,y) dx dy.$

Definition

If X and Y are continuous with joint CDF $F_{X,Y}$, their joint PDF is the derivation of the *joint CDF* with respect to x and y :

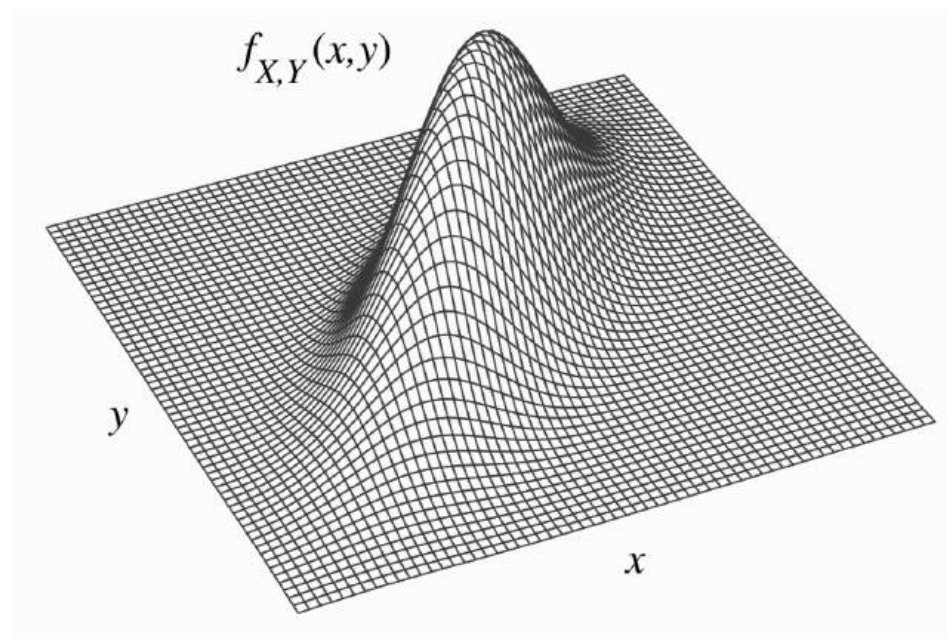
$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y).$$

③ Generally, $(x,y) \in B. \quad B \subseteq \mathbb{R}^2.$

$$P((x,y) \in B) = \iint_B f_{X,Y}(x,y) dx dy$$

Joint PDF

$$1-D: \delta \approx 0. \quad P(a \leq X \leq a + \delta) = \int_a^{a+\delta} f(x) dx \approx f_X(a) \cdot \delta.$$



$$2-D: \delta_1 \approx 0, \delta_2 \approx 0.$$

$$P(a \leq X \leq a + \delta_1, b \leq Y \leq b + \delta_2) = \int_a^{a+\delta_1} \int_b^{b+\delta_2} f_{X,Y}(x,y) dx dy \approx f_{X,Y}(a,b) \cdot \delta_1 \cdot \delta_2.$$

Marginal PDF

Definition

For continuous r.v.s X and Y with joint PDF $f_{X,Y}$, the marginal PDF of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

This is the PDF of X , viewing X individually rather than jointly with Y .

Conditional PDF

① Conditional PDF is also a valid PDF, given fixed x .

② $f_{Y|X}(\cdot|x)$ is a valid PDF: $1^\circ f_{Y|X}(y|x) \geq 0, \forall y$.

Definition

For continuous r.v.s X and Y with joint PDF $f_{X,Y}$, the conditional PDF of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y) \geq 0}{f_X(x) > 0}$$

$$\int_{-\infty}^{+\infty} f_{Y|X}(y|x) dy = \frac{\int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy}{f_X(x)} = \frac{f_X(x)}{f_X(x)} = 1.$$

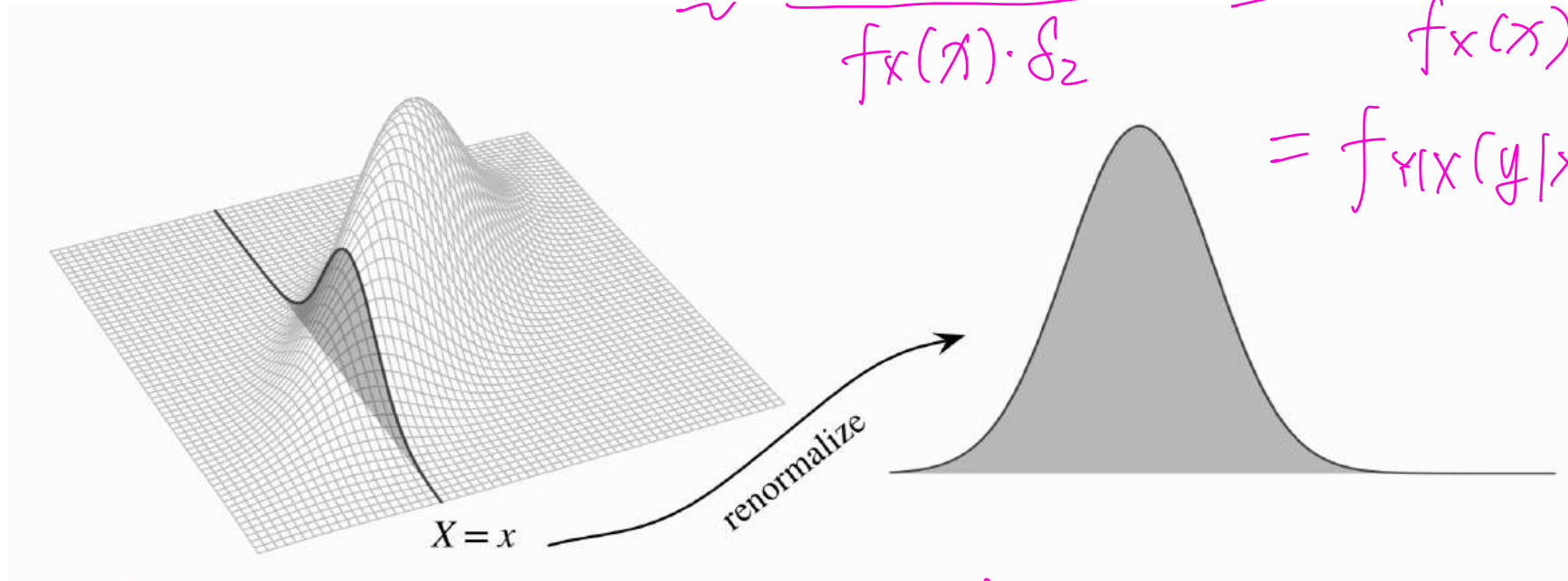
$$\delta_1, \delta_2 \approx 0$$

Conditional PDF

$$\textcircled{1} P(y \leq Y \leq y + \delta_1 | x \leq X \leq x + \delta_2) = \frac{P(y \leq Y \leq y + \delta_1, x \leq X \leq x + \delta_2)}{P(x \leq X \leq x + \delta_2)}$$

$$\approx \frac{f_{X,Y}(x, y) \cdot \delta_1 \cdot \delta_2}{f_X(x) \cdot \delta_2} = \frac{f_{X,Y}(x, y) \cdot \delta_1}{f_X(x)}$$

$$= f_{Y|X}(y|x) \delta_1$$



$$\textcircled{2} \text{ let } \delta_2 \rightarrow 0. \quad P(y \leq Y \leq y + \delta_1 | X=x) = f_{Y|X}(y|x) \cdot \delta_1$$

$$\textcircled{3} A \subseteq \mathbb{R}, \quad P(Y \in A | X=x) = \int_A f_{Y|X}(y|x) dy$$

Technique Issue

- What is the meaning of conditioning on zero-probability event $X = x$ for a continuous r.v. X .
- We are actually conditioning on the event that X falls within a small interval of x : $X \in (x - \epsilon, x + \epsilon)$ and then taking a limit as $\epsilon \rightarrow 0$.

Example

① For $0 < x < 1, 0 < y < 1$, we have

$$f_{X|Y}(x, y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{f(x, y)}{\int_0^1 f(x, y) dx} = \frac{\frac{12x(2-x-y)}{5}}{\int_0^1 \frac{12x(2-x-y)}{5} dx}$$

The joint PDF of X and Y is given by

$$f(x, y) = \begin{cases} \frac{12x(2-x-y)}{5} & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Compute the conditional PDF of X given that $Y = y$, where $0 < y < 1$.

$$= \frac{x(2-x-y)}{\int_0^1 x(2-x-y) dx} = \frac{6x(2-x-y)}{4-3y}$$

Example

① Conditional PDF, $0 < x < \infty, 0 < y < \infty$.

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f(x,y)}{\int_0^{\infty} f(x,y) dx} = \frac{\frac{e^{-x/y-y}}{y}}{\int_0^{\infty} \frac{e^{-x/y-y}}{y} dx} = \frac{1}{y} e^{-x/y}$$

Suppose that the joint PDF of X and Y is given by

$$f(x,y) = \begin{cases} \frac{e^{-x/y-y}}{y} & \text{if } 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

find $P\{X > 1 | Y = y\}$.

$$\textcircled{2} P(X > 1 | Y = y) = \int_1^{\infty} f_{X|Y}(x|y) dx = \int_1^{\infty} \frac{1}{y} e^{-x/y} dx = e^{-1/y}$$

Continuous form of Bayes' Rule and LOTP

① By definition of conditional PDF: $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$
 $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$

Theorem

For continuous r.v.s X and Y ,

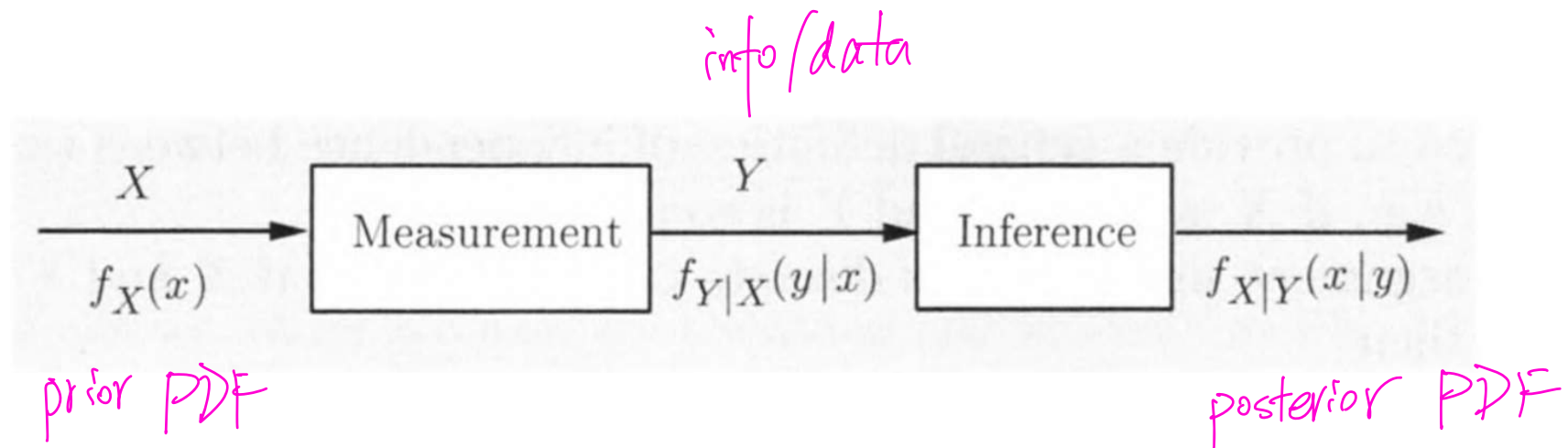
$$\textcircled{1} f_{Y|X}(y|x) = \frac{f_{Y|X}(x|y)f_Y(y)}{f_X(x)}$$

$$\textcircled{2} f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dy$$

$\downarrow$
 $f_{X,Y}(x,y)$

Proof

Bayes' Rule: Inference Perspective



Example

① r.v. λ , $f_{\lambda}(\lambda) = 2$, $1 \leq \lambda \leq \frac{3}{2}$.

② Conditional PDF: $f_{\lambda|Y}(\lambda|y) = \frac{f_{\lambda}(\lambda) \cdot f_{Y|\lambda}(y|\lambda)}{f_Y(y)} = \frac{2 \cdot \lambda e^{-\lambda y}}{\int_1^{\frac{3}{2}} 2t e^{-ty} dt}$

A light bulb produced by the GE company is known to have an exponentially distributed lifetime Y . However, the company has been experiencing quality control problems. On any given day, the parameter λ of the PDF of Y is actually a random variable. Uniformly distributed in the interval $[1, 3/2]$. We test a light bulb and record its lifetime. What we can say about the underlying parameter λ ?

$$f_{\lambda|Y}(\lambda|y) = \frac{2\lambda e^{-\lambda y}}{\int_1^{\frac{3}{2}} 2t e^{-ty} dt}$$

③ $f_Y(y) = \int_{-\infty}^{\infty} f_{\lambda}(t) f_{Y|\lambda}(y|t) dt$. LOTP.

General Bayes' Rule

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$
X continuous	$P(Y = y X = x) = \frac{f_{X Y}(x y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

Proof

① X discrete, Y continuous, $f_{Y|X}(y|X=x)$.

$$P[Y \in (y-\varepsilon, y+\varepsilon) | X=x] = \frac{P[X=x | Y \in (y-\varepsilon, y+\varepsilon)] \cdot P[Y \in (y-\varepsilon, y+\varepsilon)]}{P(X=x)}.$$

$$f_{Y|X}(y|X=x) = \lim_{\varepsilon \rightarrow 0} \frac{P[Y \in (y-\varepsilon, y+\varepsilon) | X=x]}{2\varepsilon}$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{P[X=x | Y \in (y-\varepsilon, y+\varepsilon)] \cdot \frac{P[Y \in (y-\varepsilon, y+\varepsilon)]}{2\varepsilon}}{P(X=x)}$$

$$= \frac{P(X=x | Y=y) \cdot f_Y(y)}{P(X=x)}$$

Proof

② $P(Y=y|X=x)$, Y discrete, X continuous

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \frac{P(Y=y|X \in (x-\varepsilon, x+\varepsilon))}{P(X \in (x-\varepsilon, x+\varepsilon))} &= \lim_{\varepsilon \rightarrow 0} \frac{P[X \in (x-\varepsilon, x+\varepsilon) | Y=y] P(Y=y)}{P[X \in (x-\varepsilon, x+\varepsilon)]} \\ &= \lim_{\varepsilon \rightarrow 0} \frac{\frac{P[X \in (x-\varepsilon, x+\varepsilon) | Y=y]}{2\varepsilon} \cdot P(Y=y)}{\frac{P[X \in (x-\varepsilon, x+\varepsilon)]}{2\varepsilon}} \end{aligned}$$

$$P(Y=y|X=x) = \frac{f_X(x|Y=y)}{f_X(x)} P(Y=y)$$

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① Event $\rightarrow$ prob. $\rightarrow$ conditional prob.
prior/posterior.

② Random Variable. (discrete/continuous)
Single : distribution. (Story)

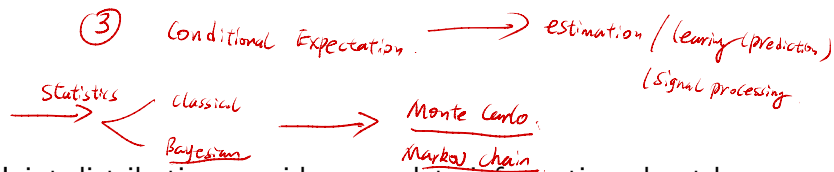


two random variables (independence)



≥ 3 r.v.s. joint distribution
Marginal distribution

Multivariate Distribution



- Joint distribution provides complete information about how multiple r.v.s interact in high-dimensional space
- Marginal distribution is the individual distribution of each r.v.
- Conditional distribution is the updated distribution for some r.v.s after observing other r.v.s

Outline

3 key themes:

① Decomposition / Integration

First-step

Conditioning

prob/expectation

① Discrete Multivariate R.V.s

② Continuous Multivariate R.V.s

③ Covariance and Correlation

④ Multinomial Distribution

⑤ Multivariate Normal

② Transformation

(invariance)

PDF

MGF

③ Approximation

Asymptotic
approximation

CLT

$n \rightarrow \infty$

Finite-size

approximation

General Bayes' Rule

$$f_X(x) \cdot 2\varepsilon \approx P(X \in (x-\varepsilon, x+\varepsilon))$$

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X=x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

$f_{Y|X}(y|x)$: Discriminative AI
 (Class label $\uparrow$ feature)
 $f_{X,Y}(x,y)$: Generative AI

Proof

$X_1, \dots, X_n$

Stochastic processes

Stochastic
information
of $X_1, \dots, X_n$

Markov : $\rightarrow$ Markov chain

Martingale : $\rightarrow$ Martingale

$\rightarrow$ branching processes

$\rightarrow$ Poisson process

General LOTP

$$(2) : \lim_{\varepsilon \rightarrow 0} \frac{P(X \in (x-\varepsilon, x+\varepsilon))}{2\varepsilon} = \sum_y \lim_{\varepsilon \rightarrow 0} \frac{P(X \in (x-\varepsilon, x+\varepsilon) | Y=y)}{2\varepsilon} \cdot P(Y=y)$$

$$f_X(x) = \sum_y f_{X|Y}(x|y) \cdot P(Y=y)$$

Y discrete

Y continuous

X discrete

$$P(X=x) = \sum_y P(X=x|Y=y)P(Y=y)$$

$$P(X=x) = \int_{-\infty}^{\infty} P(X=x|Y=y)f_Y(y)dy \quad (1)$$

X continuous

$$f_X(x) = \sum_y f_{X|Y}(x|y)P(Y=y) \quad (2) \checkmark$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dy$$

$$f_X(x) \cdot 2\varepsilon$$

Proof

$$\textcircled{1} \quad P(X=x) = \int_{-\infty}^{+\infty} \underbrace{P(X=x|Y=y)}_{\text{wavy line}} f_Y(y) dy$$

$\swarrow$
 X : discrete, Y : continuous

$$P(X=x|Y=y) = \frac{f_{Y(y|X=x)}}{f_Y(y)} \cdot P(X=x)$$

$$\Rightarrow P(X=x|Y=y) \cdot f_Y(y) = f_{Y(y|X=x)} \cdot P(X=x)$$

$$\begin{aligned} \Rightarrow \int_{-\infty}^{+\infty} P(X=x|Y=y) f_Y(y) dy &= \int_{-\infty}^{+\infty} f_{Y(y|X=x)} \cdot \underbrace{P(X=x)}_{\text{wavy line}} dy \\ &= P(X=x) \cdot \int_{-\infty}^{+\infty} f_{Y(y|X=x)} dy \\ &= \underline{P(X=x) \cdot 1} = P(X=x) \end{aligned}$$

Example

① $Y = N + S$; conditioning on $S = s$ ($s = 1$ or -1)

$$\textcircled{2} \quad \underset{\substack{\text{discrete} \\ \uparrow}}{P(S=1|Y=y)} = \frac{\underset{\substack{\text{continuous} \\ \uparrow}}{f_{Y|S}(y|1)}}{f_Y(y)} \cdot P(S=1) = \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2} \cdot p}{f_Y(y)}$$

A binary signal S is transmitted, and we are given that $P(S=1) = p$ and $P(S=-1) = 1-p$. The received signal is $Y = N + S$, where N is normal noise, with zero mean and unit variance, independent of S . What is the probability that $S=1$, as a function of the observed value y of Y ?

$$\textcircled{3} \quad f_Y(y) = f_{Y|S}(y|1) \cdot P(S=1) + f_{Y|S}(y|-1) \cdot P(S=-1)$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2} \cdot p + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2} \cdot (1-p)$$

$P(S=1) = p$ prior

$$\textcircled{4} \quad P(S=1|Y=y) = \frac{p \cdot e^y}{p \cdot e^y + (1-p) \cdot e^{-y}} = \frac{p}{p + (1-p)e^{-2y}} = \begin{cases} > p \\ = p \\ < p \end{cases}$$

posterior

$y > 0$
 $e^{-2y} < 1$
 $p < (1-p)e^{-2y}$
 $< p < 1-p \Rightarrow$

$y > 0$
 $y = 0$
 $y < 0$

Example: Comparing Exponentials of Different Rates

$$\textcircled{1} \underline{P(T_1 < T_2)} \stackrel{\text{LOTP}}{=} \int_0^\infty P(T_1 < T_2 | T_2 = t) \cdot \underline{f_{T_2}(t)} \cdot dt$$

$$\textcircled{2} P(T_1 = T_2) = 0 \quad = \int_0^\infty P(T_1 < t | T_2 = t) \cdot \underline{\lambda_2 e^{-\lambda_2 t}} dt$$

$$\underline{P(T_1 < T_2)} = \underline{P(T_1 = \min(T_1, T_2))} = \int_0^\infty \underline{P(T_1 < t)} \cdot \underline{\lambda_2 e^{-\lambda_2 t}} dt$$

Let $T_1 \sim \text{Expo}(\lambda_1)$, $T_2 \sim \text{Expo}(\lambda_2)$, T_1 and T_2 are independent.
Find $\underline{P(T_1 < T_2)}$.

$$\begin{aligned} & \left. \begin{array}{l} T_1, \dots, T_n \mid \text{Expo}(\lambda_1), \dots, \text{Expo}(\lambda_n) \\ \text{independent.} \end{array} \right\} \begin{aligned} &= \int_0^\infty (1 - e^{-\lambda_1 t}) \cdot \lambda_2 e^{-\lambda_2 t} dt \\ &= \frac{\int_0^\infty \lambda_2 e^{-\lambda_2 t} dt}{1} - \frac{\int_0^\infty \lambda_2 e^{-(\lambda_1 + \lambda_2)t} dt}{1} \\ &= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{\lambda_1}{\lambda_1 + \lambda_2} \end{aligned} \end{aligned}$$

$$P(T_2 = \min(T_1, \dots, T_n)) = \frac{\lambda_2}{\lambda_1 + \dots + \lambda_n}$$

$$\min(T_1, \dots, T_n) \sim \text{Expo}(\lambda_1 + \dots + \lambda_n)$$

Independence of Continuous R.V.s

Definition

Random variables X and Y are *independent* if for all x and y ,

$$F_{X,Y}(x,y) = F_X(x) F_Y(y)$$

If X and Y are continuous with joint PDF $f_{X,Y}$, this is equivalent to the condition

$$\underline{f_{X,Y}(x,y) = f_X(x)f_Y(y)}$$

for all x and y , and it is also equivalent to the condition

$$f_{Y|X}(y|x) = f_Y(y)$$

for all y and all x such that $f_X(x) > 0$.

Proposition

$$f_{X,Y}(x,y) = \begin{cases} 8xy & ; 0 \leq x \leq y \leq 1 \\ 0 & ; \text{otherwise.} \end{cases}$$

$\neq f_X(x) f_Y(y)$

(domain of (x,y) couple)

Theorem

$$f_X(x) = \begin{cases} 4x(1-x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}, f_Y(y) = \begin{cases} 4y^3 & , 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that the joint PDF $f_{X,Y}$ of X and Y factors as

pre-decouple

$$f_{X,Y}(x,y) = \underline{g(x)} \underline{h(y)}$$

(domain of x and y) $(-\infty, +\infty)$

for all x and y , where g and h are nonnegative functions. Then X and Y are independent. Also, if either g or h is a valid PDF, then the other one is a valid PDF too and g and h are the marginal PDFs of X and Y , respectively. (The analogous result in the discrete case also holds.)

Proof

$$f_{X,Y}(x,y) = g(x)h(y) = \underline{c \cdot g(x) \cdot \frac{1}{c} h(y)}$$

$c > 0$ is a constant.

$$\textcircled{1} \text{ Let } c = \int_{-\infty}^{\infty} h(y) dy \Rightarrow \underline{1 = \int_{-\infty}^{\infty} \frac{1}{c} h(y) dy}$$

Renormalization
key idea:

$$\begin{aligned} \Rightarrow \underline{f_X(x)} &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_{-\infty}^{\infty} \underline{c g(x)} \cdot \frac{1}{c} h(y) dy \\ &= c g(x) \cdot \underbrace{\int_{-\infty}^{\infty} \frac{1}{c} h(y) dy}_1 = \underline{c g(x)}. \end{aligned}$$

if g is a valid PDF,
 $\int_{-\infty}^{\infty} g(x) dx = 1$

$$\Rightarrow c = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} h(y) dy = 1$$

$\Rightarrow h$ is a valid PDF.

$$\textcircled{2} \int_{-\infty}^{\infty} f_X(x) dx = 1 \Rightarrow \underline{\int_{-\infty}^{\infty} c g(x) dx = 1}$$

$$\begin{aligned} \textcircled{3} \underline{f_Y(y)} &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_{-\infty}^{\infty} c g(x) \cdot \frac{h(y)}{c} dx \\ &= \frac{1}{c} h(y) \cdot \underbrace{\int_{-\infty}^{\infty} c g(x) dx}_1 = \underline{\frac{1}{c} h(y)}. \end{aligned}$$

$$\textcircled{4} f_{X,Y}(x,y) = \underline{c g(x)} \cdot \underline{\frac{1}{c} h(y)} = f_X(x) \cdot f_Y(y) \Rightarrow X, Y \text{ are independent.}$$

2D LOTUS

No LOTUS: $(X, Y) \rightarrow g(X, Y) \rightarrow E[g(X, Y)]$

LOTUS: $(X, Y) \rightarrow E[g(X, Y)]$

Theorem

Let g be a function from $\mathbb{R}^2$ to $\mathbb{R}$. If X and Y are discrete, then

$$E(g(X, Y)) = \sum_x \sum_y \underline{g(x, y)} \underline{P(X = x, Y = y)}.$$

If X and Y are continuous with joint PDF $f_{X, Y}$, then

$$E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underline{g(x, y) f_{X, Y}(x, y)} dx dy.$$

Expected Distance between Two Uniforms

$$\begin{aligned} \textcircled{1} \quad E(|X-Y|) &\stackrel{\text{Lotus}}{=} \int_0^1 \int_0^1 |x-y| f_X(x) f_Y(y) dy = \int_0^1 \int_0^1 |x-y| dx dy \\ &= \int_0^1 \int_y^1 (x-y) dx dy + \int_0^1 \int_0^y (y-x) dx dy = \underline{\underline{\frac{1}{3}}} \end{aligned}$$

$\textcircled{2}$ For $X, Y \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1)$, find $E(|X - Y|)$, $E(\max(X, Y))$, and $E(\min(X, Y))$.

$$M = \max(X, Y) \quad ; \quad L = \min(X, Y) ;$$

$$\underline{M + L = X + Y}$$

$$\Rightarrow \underline{E(M+L)} = \underline{E(X+Y)} \Rightarrow \underline{E(M) + E(L)} = E(X) + E(Y) = \frac{1}{2} + \frac{1}{2} = \underline{1}$$

$$\textcircled{3} \quad M - L = \max(X, Y) - \min(X, Y) = \begin{cases} X - Y & \text{if } X \geq Y \\ Y - X & \text{if } X < Y \end{cases} = |X - Y|$$

$$\Rightarrow \underline{E(M-L)} = E(|X-Y|) = \underline{\frac{1}{3}}$$

$$\underline{E(M) - E(L)} = \underline{\frac{1}{3}} \quad \Rightarrow \textcircled{4} \quad E(M) = \frac{2}{3} ; E(L) = \underline{\underline{\frac{1}{3}}}$$

Expected Distance between Two Normals

① Method 1: $E(|X-Y|) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |x-y| \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dx dy$

② Method 2: $Y \sim N(0,1)$; $-Y \sim N(0,1)$.

$X-Y \sim N(0,2)$

$X-Y = \sqrt{2}Z$, $Z \sim N(0,1)$

For $X, Y \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, find $E(|X-Y|)$.

$\Rightarrow E(|X-Y|) = E(|\sqrt{2}Z|) = \sqrt{2} E(|Z|)$.

$$\begin{aligned} E(|Z|) &= \int_{-\infty}^{\infty} |z| \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz = 2 \int_0^{\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\ &= \sqrt{\frac{2}{\pi}} \end{aligned}$$

$\Rightarrow E(|X-Y|) = \frac{2}{\sqrt{\pi}}$

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- 1 Discrete Multivariate R.V.s
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Covariance

Definition

The covariance between r.v.s X and Y is

$$\text{Cov}(X, Y) = E((X - EX)(Y - EY)).$$

Multiplying this out and using linearity, we have an equivalent expression:

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

Key Properties of Covariance

① • $\text{Cov}(X, X) = \text{Var}(X)$.

② • $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.

• $\text{Cov}(X, c) = 0$ for any constant c .

• $\text{Cov}(a \cdot X, Y) = a \cdot \text{Cov}(X, Y)$ for any constant a .

• $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.

④ • $\text{Cov}(X + Y, Z + W) =$
 $\text{Cov}(X, Z) + \text{Cov}(X, W) + \text{Cov}(Y, Z) + \text{Cov}(Y, W)$.

• $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. *proof*

• For n r.v.s $X_1, \dots, X_n$,

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

$$\text{Var}(X+Y) \stackrel{①}{=} \text{Cov}(X+Y, X+Y)$$

$$\stackrel{④}{=} \text{Cov}(X, X) + \text{Cov}(X, Y)$$

$$+ \text{Cov}(Y, X) + \text{Cov}(Y, Y)$$

$$\stackrel{①}{=} \text{Var}(X) + \text{Var}(Y)$$

$$+ \text{Cov}(X, Y)$$

$$+ \text{Cov}(Y, X)$$

$$\stackrel{②}{=} \text{Var}(X) + \text{Var}(Y)$$

$$+ 2\text{Cov}(X, Y)$$

Proof

Correlation

Definition

Linear Correlation

The correlation between r.v.s X and Y is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

(This is undefined in the degenerate cases $\text{Var}(X) = 0$ or $\text{Var}(Y) = 0$.)

Definition

Given r.v.s X and Y , if $\text{Cov}(X, Y) = 0$ or $\text{Corr}(X, Y) = 0$, X and Y are uncorrelated.

Uncorrelated

X, Y are independent.

$$\text{COV}(X, Y) = E[(X - E X)(Y - E Y)]$$

$$= \underbrace{E[X - E X]}_0 \cdot \underbrace{E[Y - E Y]}_0 = 0$$

Theorem

If X and Y are independent, then they are uncorrelated.

Uncorrelated $\nRightarrow$ Independent

$$\text{Cov}(X, Y) = \frac{E(XY)}{0} - \frac{E(X)E(Y)}{0} = 0$$

Example: $X \sim N(0, 1)$; $Y = X^2$

1^o. $E(X) = 0$; $\Rightarrow E(X) \cdot E(Y) = 0$

2^o. $E(XY) = E(X^3) = 0$

3^o $\Rightarrow \text{Cov}(X, Y) = 0$

X, Y are linear uncorrelated (potentially nonlinear correlated)
but, X, Y are dependent.

$$E(X^3) = \int_{-\infty}^{\infty} x^3 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 0$$

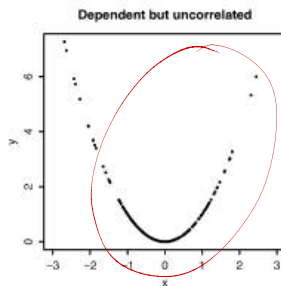
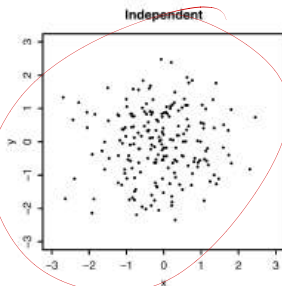
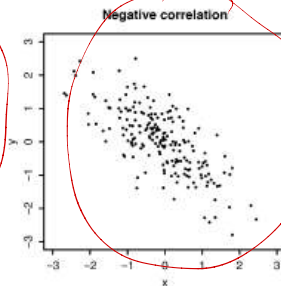
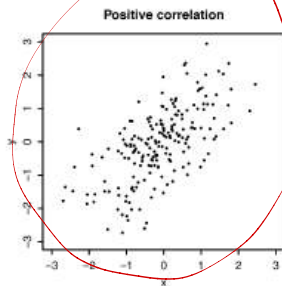
$$g(-x) = -g(x)$$

$$\int_{-\infty}^{\infty} g(x) dx = \int_0^{\infty} g(x) dx + \int_{-\infty}^0 g(x) dx = \int_0^{\infty} g(x) dx - \int_0^{\infty} g(x) dx = 0$$

Covariance & Correlation

- Measure a tendency of two r.v.s X & Y to go up or down together
- Positive covariance (Correlation): when X goes up, Y also tends to go up
- Negative covariance (Correlation): when X goes up, Y tends to go down

Correlation



Correlation Bounds

Cauchy - Schwarz Inequality

$$|E[XY]| \leq \sqrt{E[X^2] \cdot E[Y^2]}$$

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \sum_{i=1}^n a_i^2 \cdot \sum_{i=1}^n b_i^2$$

$$1^0. \quad f(t) = E[(X - tY)^2] = E[X^2 - 2tXY + t^2Y^2]$$

$$= E[X^2] - 2tE[XY] + t^2E[Y^2]$$

$$= t^2E[Y^2] - 2tE[XY] + E[X^2]$$

$$\left(\int_a^b f(x)g(x)dx\right)^2 \leq \int_a^b f(x)^2 dx \cdot \int_a^b g(x)^2 dx$$

Theorem

$$\Delta = (2E[XY])^2 - 4 \cdot E[X^2] \cdot E[Y^2] \leq 0 \Rightarrow E[XY]^2 \leq E[X^2] \cdot E[Y^2]$$

For any r.v.s X and Y ,

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

$$2^0. \quad X - E[X], Y - E[Y] \xrightarrow{\text{C.S.}} E[(X - E[X])(Y - E[Y])] \leq \sqrt{E[(X - E[X])^2] \cdot E[(Y - E[Y])^2]}$$

$$\Rightarrow \text{Cov}^2(X, Y) \leq \text{Var}(X) \cdot \text{Var}(Y)$$

$$\vec{a} \cdot \vec{b} \leq \|\vec{a}\| \cdot \|\vec{b}\|$$

$$\left| \cos \theta \right| = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}$$

$\Rightarrow$

$$\frac{\text{Cov}^2(X, Y)}{\text{Var}(X) \cdot \text{Var}(Y)} \leq 1$$

$$\Rightarrow \text{Corr}^2(X, Y) \leq 1$$

≤ 1

$$\Rightarrow -1 \leq \text{Corr}(X, Y) \leq 1$$

Outline

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Multinomial Distribution
- 5 Multivariate Normal

Story

$k=2 \rightarrow \text{Binomial.}$

Each of n objects is independently placed into one of k categories. An object is placed into category j with probability p_j , where the p_j are nonnegative and $\sum_{j=1}^k p_j = 1$. Let X_1 be the number of objects in category 1, X_2 the number of objects in category 2, etc., so that $X_1 + \dots + X_k = n$. Then $\mathbf{X} = (X_1, \dots, X_k)$ is said to have the Multinomial distribution with parameters n and $\mathbf{p} = (p_1, \dots, p_k)$. We write this as $\mathbf{X} \sim \text{Mult}_k(n, \mathbf{p})$.

$$\mathbf{p} = (p_1, p_2) \quad (p_1, 1-p_1)$$
$$p_1 + p_2 = 1$$
$$p_1 \neq$$

Multinomial Joint PMF

Method 1^o: $\binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdot \binom{n-n_1-n_2}{n_3} \cdots \binom{n-n_1-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1! \cdots n_k!}$

Method 2^o: $\frac{n!}{n_1! \cdot n_2! \cdots n_k!}$

Theorem

If $\mathbf{X} \sim \text{Mult}_k(n, \mathbf{p})$, then the joint PMF of $\mathbf{X}$ is

$$P(X_1 = n_1, \dots, X_k = n_k) = \frac{n!}{n_1! n_2! \cdots n_k!} \cdot p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

for $n_1, \dots, n_k$ satisfying $n_1 + \cdots + n_k = n$.

Proof

Multinomial Marginals

Successful event: object $\rightarrow j$ -th category.

Theorem

If $\mathbf{X} \sim \text{Mult}_k(n, \mathbf{p})$, then $X_j \sim \text{Bin}(n, p_j)$.

Multinomial Lumping

event $A =$ "fall into i^{th} category".

$B =$ "----- j^{th} -----".

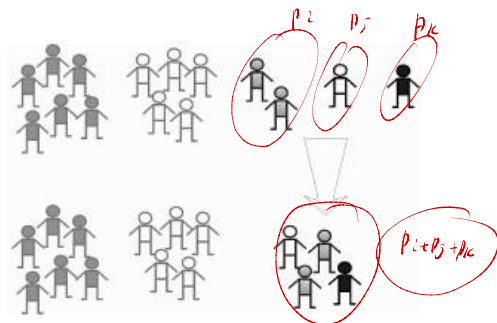
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \left. \begin{array}{l} P(A \cap B) = 0 \\ \end{array} \right\}$$
$$= P(A) + P(B) = p_i + p_j$$

Theorem

If $\mathbf{X} \sim \text{Mult}_k(n, \mathbf{p})$, then for any distinct i and j ,
 $X_i + X_j \sim \text{Bin}(n, p_i + p_j)$. The random vector of counts obtained
from merging categories i and j is still Multinomial. For example,
merging categories 1 and 2 gives

$$(X_1 + X_2, X_3, \dots, X_k) \sim \text{Mult}_{k-1}(n, (p_1 + p_2, p_3, \dots, p_n)).$$

Multinomial ~~Lumping~~



Multinomial Conditioning

1°. Given n_1 objects in Category 1, the remaining $n - n_1$ objects

landing into categories 2, ..., k: $\text{Mult}_{k-1}(n - n_1, \underline{p'})$

Theorem

If $\mathbf{X} \sim \text{Mult}_k(n, \mathbf{p})$, then

$$\frac{p_j'}{1 - p_1} = \text{Prob} \left(\underbrace{\text{landing into category } j}_A \mid \underbrace{\text{not landing into Category 1}}_B \right)$$

$$(X_2, \dots, X_k) \mid \underline{X_1 = n_1} \sim \text{Mult}_{k-1}(n - n_1, (\underline{p'_2}, \dots, \underline{p'_k})),$$

where $p'_j = p_j / (p_2 + \dots + p_k)$.

$$p_1 + p_2 + \dots + p_k = 1$$

$$p'_2 + \dots + p'_k = 1$$

renormalization

$$= \text{Prob. (Landing into Category } j)$$

$$\text{Prob. (not Landing into Category 1)}$$

$$= \frac{p_j}{1 - p_1} = \frac{p_j}{p_2 + \dots + p_k} = \underline{p'_j}$$

Covariance in A Multinomial

$$Z \sim \text{Bin}(n, p) ; \text{Var}(Z) = np(1-p)$$

1°. w.l.o.g. $i=1 ; j=2$.

$$X_1 \sim \text{Bin}(n, p_1)$$

$$X_2 \sim \text{Bin}(n, p_2)$$

$$X_1 + X_2 \sim \text{Bin}(n, p_1 + p_2)$$

Theorem

Let $(X_1, \dots, X_k) \sim \text{Mult}_k(n, \mathbf{p})$, where $\mathbf{p} = (p_1, \dots, p_k)$. For $i \neq j$,
 $\text{Cov}(X_i, X_j) = -np_i p_j < 0$

$$2^\circ. \quad \text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$$

$$n(p_1 + p_2)(1 - p_1 - p_2) = np_1(1 - p_1) + np_2(1 - p_2) + 2\text{Cov}(X_1, X_2)$$

$$\Rightarrow \text{Cov}(X_1, X_2) = -np_1 p_2 < 0$$

Proof

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Multivariate Normal Distribution

Definition

A random vector $\mathbf{X} = (X_1, \dots, X_k)$ is said to have a Multivariate Normal (MVN) distribution if every linear combination of the X_j has a Normal distribution. That is, we require

$$\underline{t_1 X_1 + \dots + t_k X_k}$$

to have a Normal distribution for any choice of constants $t_1, \dots, t_k$. If $t_1 X_1 + \dots + t_k X_k$ is a constant (such as when all $t_i = 0$), we consider it to have a Normal distribution, albeit a degenerate Normal with variance 0. An important special case is $k = 2$; this distribution is called the Bivariate Normal (BVN).

Non-example of MVN

$$X = (X_1, \dots, X_k) \text{ if } X_j \sim N(\mu_j, \sigma_j^2)$$

$$X \neq \text{MVN}$$

$$(1) X \sim N(0, 1), \quad S = \begin{cases} 1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$$

S and X are independent.

$$(2) Y = S \cdot X \sim N(0, 1) : \quad P(Y \leq y) = P(S \cdot X \leq y)$$

$$(3) P(X+Y=0)$$

$$= P(S = -1) = \frac{1}{2}$$

$X+Y$ is NOT continuous.
 $\Rightarrow$ Normal.

$$(X, Y) \neq \text{MVN}$$

$$\begin{aligned} &= \underbrace{P(SX \leq y | S=1)}_{\text{L.O.P.}} \cdot P(S=1) \\ &\quad + \underbrace{P(SX \leq y | S=-1)} \cdot P(S=-1) \\ &= \underbrace{P(X \leq y | S=1)} \cdot \frac{1}{2} + \underbrace{P(-X \leq y | S=-1)} \cdot \frac{1}{2} \\ &= P(X \leq y) \cdot \frac{1}{2} + \frac{P(-X \leq y)}{P(X \geq -y)} \cdot \frac{1}{2} \\ &= \underbrace{P(X \leq y)}_{\Rightarrow Y \sim N(0, 1)} \cdot \frac{1}{2} \end{aligned}$$

Actual MVN

Sum of ^{MGF} independent Normal $\rightarrow$ Normal.

$$\textcircled{1} \quad Z, W \stackrel{\text{i.i.d.}}{\sim} N(0,1)$$

$$\textcircled{2} \quad (Z+2W, 3Z+5W) \text{ is a MVN.}$$

$$\forall t_1, t_2 \in \mathbb{R}, \quad t_1(Z+2W) + t_2(3Z+5W)$$

$$= (t_1+3t_2)\underline{Z} + (2t_1+5t_2)\underline{W}$$

$$\sim \text{Normal.}$$

Theorem

$$(X_1, X_2, X_3) \text{ MVN} \rightarrow \exists t_1, t_2, t_3 \in \mathbb{R}.$$

$$t_1 X_1 + t_2 X_2 + t_3 X_3 \sim \text{Normal}.$$

Theorem

$$\text{Let } t_3 = 0, \exists t_1, t_2 \in \mathbb{R}.$$

If (X_1, X_2, X_3) is Multivariate Normal, then so is the subvector (X_1, X_2) .

Theorem

Theorem

If $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{Y} = (Y_1, \dots, Y_m)$ are MVN vectors with $\mathbf{X}$ independent of $\mathbf{Y}$, then the concatenated random vector $\mathbf{W} = (X_1, \dots, X_n, Y_1, \dots, Y_m)$ is Multivariate Normal.

Parameters of MVN

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(x^2+y^2-2\rho xy)\right\} \quad \checkmark$$

Joint pdf

$$P20 \quad f_{X,Y}(x,y) = \underbrace{h(x)}_{\text{marginal}} \underbrace{g(y)}_{\text{marginal}}$$

X and Y are independent!

$$= \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp\left\{-\frac{1}{2}(x,y)\Sigma^{-1}(x,y)^T\right\}$$

Parameters of an MVN random vector $(X_1, \dots, X_k)$ are:

- the mean vector $(\mu_1, \dots, \mu_k)$, where $E(X_j) = \mu_j$.
- the covariance matrix, which is the $k \times k$ matrix of covariance between components, arranged so that the row i , column j entry is $\text{Cov}(X_i, X_j)$.

Standard Bivariate Normal, (X,Y)

$$X \sim N(0,1)$$

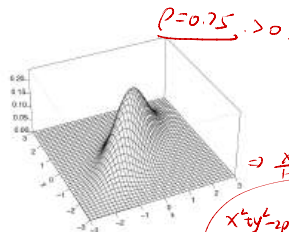
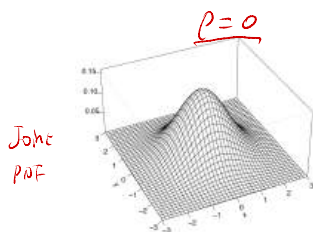
$$Y \sim N(0,1)$$

$$\text{Corr}(X,Y) = \rho \in (-1,1)$$

mean vector $(0,0)$

covariance matrix $\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X,Y) \\ \text{Cov}(X,Y) & \text{Var}(Y) \end{pmatrix} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$

Joint PDF of Bivariate Normal Distributions



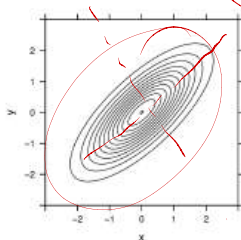
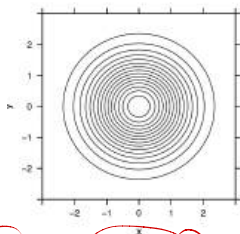
$$X_1 = \frac{1}{\sqrt{2}}(x+y)$$

$$Y_1 = \frac{1}{\sqrt{2}}(x-y)$$

$$\Rightarrow \frac{X_1^2}{1-\rho} + \frac{Y_1^2}{1-\rho} = C > 0$$

Contour

$$\frac{X^2 + Y^2 - 2\rho xy}{1-\rho^2} = C > 0$$



$\rho > 0$ $\downarrow$

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} = 1$$

$\rho > 0$; $X^2 + Y^2 = C > 0$

Joint MGF

$$\textcircled{1} W \sim \text{Normal}. \quad \text{MGF} = e^{\underline{E(W)} + \frac{1}{2}\text{Var}(W)} \\ E[e^{tW}] = \underbrace{e^{\underline{E(W)} \cdot t + \frac{1}{2}\text{Var}(W) \cdot t^2}} \quad \checkmark$$

$$\textcircled{2} (X_1, \dots, X_k) \text{ MUV.} \Rightarrow \text{Joint MGF. } E[e^{t_1 X_1 + \dots + t_k X_k}]$$

Definition

The joint MGF of a random vector $\mathbf{X} = (X_1, \dots, X_k)$ is the function which takes a vector of constants $\mathbf{t} = (t_1, \dots, t_k)$ and returns

$$= \exp \left\{ \underline{t_1 E(X_1) + \dots + t_k E(X_k)} + \frac{1}{2} \text{Var}(t_1 X_1 + \dots + t_k X_k) \right\}$$

$$M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}}) = E(e^{\underline{t_1 X_1 + \dots + t_k X_k}}).$$

We require this expectation to be finite in a box around the origin in $\mathbb{R}^k$; otherwise we say the joint MGF does not exist.

Theorem

Theorem

Within an MVN random vector, uncorrelated implies independent.
That is, if $\mathbf{X} \sim \text{MVN}$ can be written as $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$, where $\mathbf{X}_1$ and $\mathbf{X}_2$ are subvectors, and every component of $\mathbf{X}_1$ is uncorrelated with every component of $\mathbf{X}_2$, then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent.
In particular, if (X, Y) is Bivariate Normal and $\text{Corr}(X, Y) = 0$, then X and Y are independent.

Proof

① Recall fact: $(X \sim N(\mu, \sigma^2))$, $M_X(t) \stackrel{MGF}{=} e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

② Bivariate Normal (X, Y) , $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$
 $\text{Corr}(X, Y) = \rho$.

$$\Rightarrow \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$= \text{Var}(X) + \text{Var}(Y) + 2\rho \sqrt{\text{Var}(X)\text{Var}(Y)}$$

③ Joint MGF $M_{X,Y}(s, t) = E[e^{sX+tY}] = \text{Exp}\left\{s\mu_1 + t\mu_2 + \frac{1}{2}\text{Var}(sX+tY)\right\}$
 $= \text{Exp}\left\{s\mu_1 + t\mu_2 + \frac{1}{2}(s^2\sigma_1^2 + t^2\sigma_2^2 + 2st\sigma_1\sigma_2\rho)\right\}$

④ $\rho = 0$ $\Rightarrow M_{X,Y}(s, t) = \text{Exp}\left\{s\mu_1 + t\mu_2 + \frac{1}{2}s^2\sigma_1^2 + \frac{1}{2}t^2\sigma_2^2\right\}$
unrelated $= \text{Exp}\left\{s\mu_1 + \frac{1}{2}s^2\sigma_1^2}\right\} \cdot \text{Exp}\left\{t\mu_2 + \frac{1}{2}t^2\sigma_2^2}\right\}$
 $= M_X(s) \cdot M_Y(t) \Rightarrow X \text{ and } Y \text{ are independent.}$

Bivariate Normal Generation

1°. $Z = aX + bY$, $(a, b, c, d) \in \mathbb{R}$? $\Rightarrow (Z, W)$ bivariate Normal.
 $W = cX + dY$
 $Z, W \sim \mathcal{N}(0, 1)$
 $\text{Corr}(Z, W) = \rho$

2°. $E(Z) = E(aX + bY) = a \underbrace{E(X)}_0 + b \underbrace{E(Y)}_0 = 0$;
 $E(W) = 0$;

Suppose that we have access to i.i.d. r.v.s $X, Y \sim \mathcal{N}(0, 1)$, but want to generate a Bivariate Normal (Z, W) with $\text{Corr}(Z, W) = \rho$ and Z, W marginally $\mathcal{N}(0, 1)$, for the purpose of running a simulation. How can we construct Z and W from linear combinations of X and

Y? 3°. $\text{Var}(Z) = \text{Var}(aX + bY) = \text{Var}(aX) + \text{Var}(bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
 $= a^2 + b^2 = 1$.

$$\text{Var}(W) = \text{Var}(cX + dY) = c^2 + d^2 = 1$$

4°. $\text{Corr}(Z, W) = \text{Cov}(Z, W) = \rho \Rightarrow \text{Cov}(aX + bY, cX + dY) = \rho$

$$\Rightarrow aC \text{Cov}(X, X) + b d \text{Cov}(X, Y) + 0 = \rho \Rightarrow \underline{ac + bd = \rho}$$

Solution

$$5^{\circ} \quad \begin{cases} a^2 + b^2 = 1 \\ c^2 + d^2 = 1 \\ \underline{ac + bd = \rho} \end{cases}$$

Find one solution is enough.

$$b=0; \quad \Rightarrow a^2=1 \quad \Rightarrow \overset{\text{pick}}{a}=1; \quad c=\rho.$$

$$\Rightarrow d^2 = 1 - \rho^2 \quad \Rightarrow \text{pick } d = \sqrt{1 - \rho^2}$$

$$6^{\circ}. \quad Z = aX + bY = \underline{X}.$$

$$W = cX + dY = \underline{\rho X + \sqrt{1 - \rho^2} Y}.$$

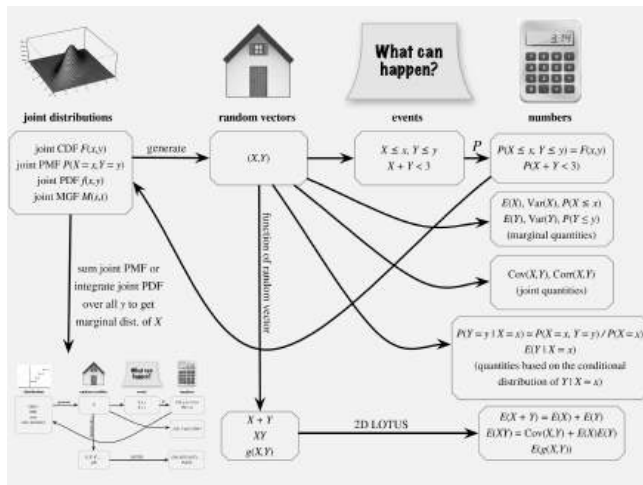
(Z, W) is the desired bivariate Normal distribution

$$\underline{\text{Joint PDF?}} \quad (Z, W) = \underline{g(X, Y)} \dots$$

Summary 1: Discrete & Continuous

	Two discrete r.v.s	Two continuous r.v.s
Joint CDF	$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$	$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$
Joint PMF/PDF	$P(X = x, Y = y)$ Joint PMF is nonnegative and sums to 1: $\sum_x \sum_y P(X = x, Y = y) = 1.$	$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$ - Joint PDF is nonnegative and integrates to 1: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1.$ - To get probability, integrate joint PDF over region of interest.
Marginal PMF/PDF	$P(X = x) = \sum_y P(X = x, Y = y)$ $= \sum_y P(X = x Y = y) P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$ $= \int_{-\infty}^{\infty} f_{X Y}(x y) f_Y(y) dy$
Conditional PMF/PDF	$P(Y = y X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$ $= \frac{P(X = x Y = y) P(Y = y)}{P(X = x)}$	$f_{Y X}(y x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$ $= \frac{f_{X Y}(x y) f_Y(y)}{f_X(x)}$
Independence	$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y)$ $P(X = x, Y = y) = P(X = x) P(Y = y)$ for all x and y. $P(Y = y X = x) = P(Y = y)$ for all x and y, $P(X = x) > 0$.	$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y)$ $f_{X,Y}(x,y) = f_X(x) f_Y(y)$ for all x and y. $f_{Y X}(y x) = f_Y(y)$ for all x and y, $f_X(x) > 0$.
LOTUS	$E(g(X,Y)) = \sum_x \sum_y g(x,y) P(X = x, Y = y)$	$E(g(X,Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y) dx dy$

Summary 2: Multivariate Distribution



References

- Chapter 7 of **BH**
- Chapters 2 & 3 & 4 of **BT**